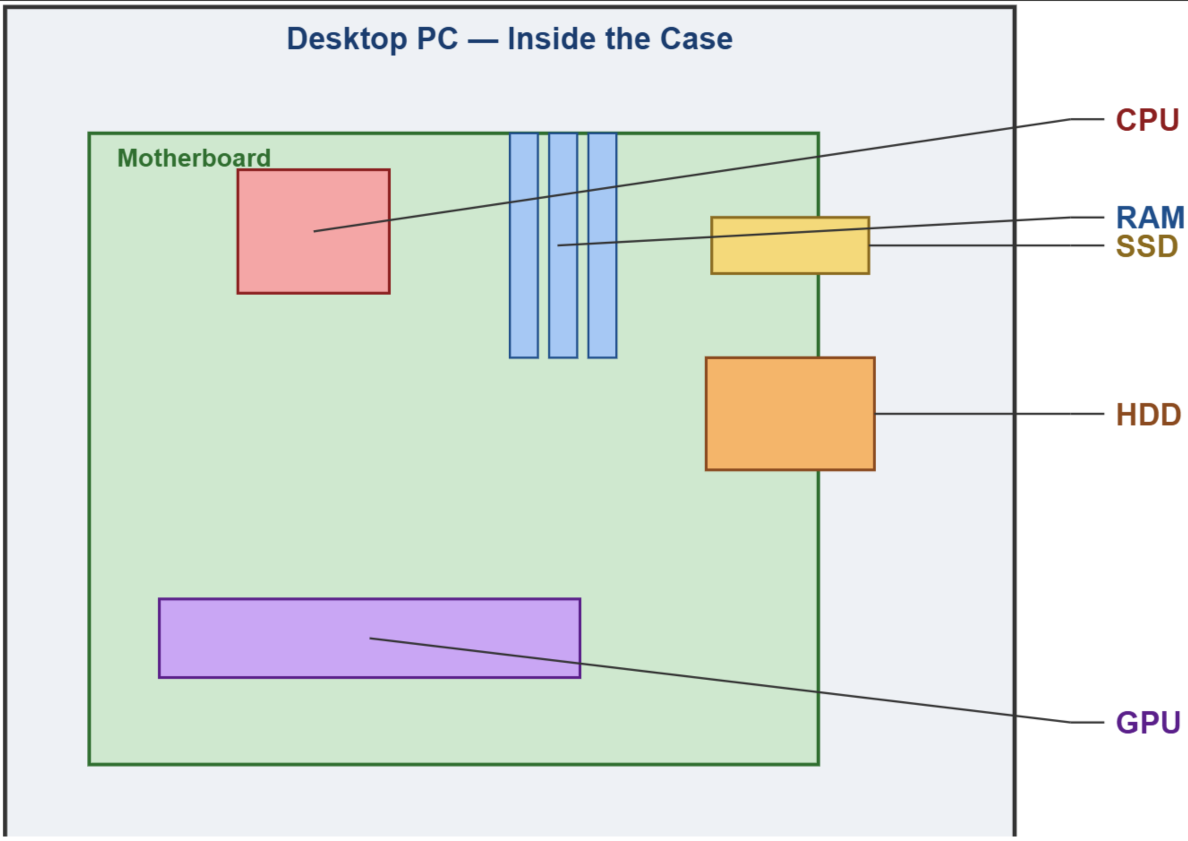
Day 1 Assignment: Overview of Hardware Components

Tasks

1.

Draw a labeled diagram of a desktop PC, clearly marking the motherboard, CPU, RAM, GPU, and both HDD and SSD storage components.



2.

**Create a comparison table showing the main differences between HDD and SSD storage in terms of speed, durability, and typical use cases.

Hint: Think about which one is faster, which is more shock-resistant, and where you would usually find each type (laptop, gaming PC, etc.).**

Feature	HDD (Hard Disk Drive)	SSD (Solid State Drive)
Speed	Slower — uses spinning magnetic disks and a moving read/write head	Much faster — uses flash memory chips with no moving parts
Durability	Less durable — moving parts can be damaged by shocks or drops	More durable / shock-resistant — no moving parts to break
Typical Use Case	Bulk storage, backups, older desktops, budget builds — cheap per GB	Boot drives, gaming PCs, laptops — faster load times, worth the higher cost
Noise & Heat	Produces some noise/vibration and more heat (spinning disk motor)	Silent — no moving parts, generally runs cooler
Price per GB	Cheaper	More expensive

3.

Write a short explanation (2-3 sentences each) describing the role of the CPU, RAM, and GPU in running a game like PUBG or Free Fire on a PC.

CPU (Processor):

The CPU handles the core game logic like physics calculations, AI behaviour of enemies, hit detection, and processing player input. In a battle royale game like PUBG or Free Fire, it also manages network data as the map, other players, and events are constantly updated in real time.

RAM (Memory):

RAM temporarily holds the game data that needs to be accessed instantly like textures, the current map area, player states, and loaded assets. Enough RAM lets the game keep this data ready without constant reloading, which prevents stutters and lag while moving around the map.

GPU (Graphics Card):

The GPU renders everything you actually see the terrain, character models, lighting, and visual effects like gunfire or explosions. A stronger GPU pushes higher frame rates and smoother visuals, which matters a lot in fast-paced shooters like PUBG or Free Fire.

4.

Imagine you want to upgrade your PC to run the latest version of FIFA smoothly. List which hardware components you would consider upgrading and explain why for each.

To run the latest FIFA smoothly at good settings and frame rates, these are the components worth upgrading:

- GPU (Graphics Card): FIFA relies heavily on real-time rendering of players, stadiums, crowds, and effects. A better GPU is usually the single biggest upgrade for higher frame rates and higher graphics settings.
- CPU (Processor): A modern game engine needs a capable CPU to handle physics (ball movement, player collisions), AI for opposing teams, and to avoid bottlenecking a strong GPU. Upgrading an old/weak CPU prevents frame drops during busy match moments.
- RAM: More RAM (e.g. going from 8GB to 16GB) helps the game load textures and assets smoothly, reduces stutter, and lets the game run comfortably alongside background apps like Discord or a stream overlay.
- SSD (Storage): Moving the game from an HDD to an SSD dramatically cuts load times (match loading, menu navigation) and makes texture streaming smoother during play.